METRIC

DOD-STD-2139(SH)
NOTICE 2
12 February 1988

MILITARY STANDARD

SHIP INFORMATION BOOK,
PREPARATION OF (METRIC)

TO THE HOLDERS OF DOD-STD-2139(SH):

1. THE FOLLOWING PAGES OF THIS STANDARD HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
cover page	28 December 1981	REPRINTED WITHOUT CHANGE	
ii	12 February 1988	ii	28 December 1981
iii/iv	12 February 1988	iii/iv	28 December 1981
1	28 December 1981	REPRINTED WITHOUT CHANGE	
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29	12 February 1988	29	28 December 1981

2. RETAIN THIS NOTICE AND INSERT BEFORE THE TABLE OF CONTENTS.

3. Holders of DOD-STD-2139(SH) will verify that page changes and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with the appended pages, is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or canceled.

Preparing activity:
Navy - SH
(Project TMSS-N210)

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28 December 1981

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DEPARTMENT OF THE NAVY
NAVAL SEA SYSTEMS COMMAND

Washington, DC 20362-5101

Ship Information Book, Preparation of (Metric)

1. This Military Standard is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Supersedes page ii of 28 December 1981

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1. SCOPE

1.1 Scope. This standard contains requirements for the preparation of Ship Information Book(s) (SIB) that is to be used as a system-level ready reference for use by operating, maintenance, and overhaul personnel, and for use in the training of such personnel in Navy schools and on board ship.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-38784 - Manuals, Technical: General Style and Format Requirements.
MIL-P-38790 - Printing Production of Technical Manuals, General Requirements for, (Photolithographic Negatives, Reproduction Assembly Sheets, Printing).

STANDARDS

MILITARY

MIL-STD-12 - Abbreviations For Use on Drawings, Specifications, Standards and in Technical Documents.
MIL-STD-15-2 - Electrical Wiring Equipment Symbols for Ships' Plans, Part 2.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

803-5001049 - Piping System, Symbols and Abbreviations.

PUBLICATIONS

MILITARY

DOD 5220.22 Series - Industrial Security Manual for Safe-Guarding Classified Information.
NAVSEA OP 1700 - Fire Control Symbols Standard, Volume 1.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
Y32.4 - Graphic Symbols for Plumbing Fixtures for Diagrams Used in Architectural and Building Construction.
Y32.10 - Fluid Power Diagrams, Graphic Symbols for.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)
F 856 - Heating, Ventilation and Air Conditioning Symbols and Abbreviations.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

3. DEFINITIONS (Not applicable)

4. GENERAL REQUIREMENTS (See appendix for data requirements)

4.1 General. The Ship Information Book (SIB) shall be prepared on an individual ship basis. It shall be complete and detailed as required so that it will serve as the primary intrasystem and intersystem technical publication covering installation and operation of ship systems and equipment. It shall provide a basic reference source, for ships' force, for planning and overhaul yards, for fleet and training commands, and for tenders and other naval activities. Coverage shall include mechanical, heating, ventilation and air conditioning, piping, electrical, and electronics systems and equipment on the ship. It shall provide a thorough understanding of the installations on the ship, together with intrasystem and intersystem capabilities and limitations. It shall contain or provide reference to technical information required by the crew as required for proper operation and maintenance of the systems and equipment covered. Document reference numbers shall be standard throughout all chapters of a volume, that is, reference (1) shall be a reference to the same document throughout the volume. Only very common acronyms shall be used, each of which should be spelled out when first utilized.

4.1.1 Current system diagrams may be used as the diagrammatics for each chapter. If used, the diagrams shall be revised to remove numbered references, revision symbols, notes that are not applicable, title blocks, and other information extraneous to the SIB. Connection diagrams for connecting major components in a system or plant and other illustrations required shall be prepared. Illustrations shall be neat and legible.

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4.1.2 The text shall not include detailed component operating and maintenance instructions which would duplicate those in the system and component technical manuals, but shall instead refer to these technical manuals for operation and maintenance. The text shall, however, include clear instructions relating component operation to the overall system operation. Reference to the Naval Ships' Technical Manual (NSTM) shall be made where appropriate.

4.2 Divisions. The SIB shall be divided into volumes, parts, chapters, sections, and paragraphs in accordance with MIL-M-38784. Volumes of the SIB shall not exceed 40 mm (1-1/2 inches) in thickness.

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4.2.1 Surface ships.

4.2.1.1 Over 90 meters (300 feet). The SIB shall consist of the following separately-bound volumes, or parts of volumes, as applicable, if listed separately below, with the titles as indicated:

(a) Volume 1 - Hull and hull mechanical systems.
(b) Volume 2 - Machinery plant.
    (1) Part 1 - Propulsion plant, general design and operating procedures.
    (2) Part 2 - Auxiliary machinery, piping, air conditioning, ventilation, and heating systems.
(c) Volume 3 - Power and lighting systems.
    (1) Part 1 - General description and design information of systems.
    (2) Part 2 - General description of electrically operated auxiliaries.
(d) Volume 4 - Electronic systems.
(e) Volume 5 - Interior communication systems.
    (1) Part 1 - Interior communication systems.
    (2) Part 2 - Sound-powered telephone systems, voice tubes, and message passing facilities.
(f) Volume 6 - Weapons control systems.
(g) Volume 7 - Ballasting systems, as applicable.

4.2.1.2 Under 90 meters (300 feet). The SIB may be bound in a single large volume, provided the titles indicated in 4.2.1.1 are clearly separated and indicated (for example, thumb-indexed), and width limitation is not exceeded.

4.2.2 Submarines. SIB shall consist of the following separately-bound volumes with titles as indicated:

(a) Volume 1 - General information,
(b) Volume 2 - Tactical facilities,
(c) Volume 3 - Ship control systems,
(d) Volume 4 - Steam and diesel propulsion plant systems,
(e) Volume 5 - Electrical power systems,
(f) Volume 6 - Ship service systems, and
(g) Volume 7 - Hull, mechanical, and ship emergency systems.

5. DETAILED REQUIREMENTS (See appendix for data requirements)

5.1 Style and format. Unless otherwise specified herein, the style and format requirements for preparation of the SIB and related materials shall be in accordance with MIL-M-38784. Abbreviations and symbols shall comply with MIL-STD-12, MIL-STD-15-2, ASTM F 856, NAVSEA Drawing 803-5001049, and ANSI Y32.4, as applicable.

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5.1.1 Format.

5.1.1.1 Numbering of pages, illustrations, and tables. Text pages, illustrations, and tables shall be numbered consecutively in Arabic numerals within each section of a chapter. Assignments shall be a three-part number; the first two numbers separated by a hyphen and the last two numbers separated by a decimal point. The first number shall represent the chapter number, the second number shall represent the section number, and the third number shall represent the order within the section. For example: The designation 3-4.12 shall be the 12th page in the 4th section of chapter 3. For volume arrangement, see 5.3.3.

5.1.1.2 Illustrations. Illustrations shall be in accordance with 5.1, but shall be microform-compatible, as specified in 5.1.1.4. Graphic symbols for fluid power diagrams shall be in accordance with ANSI Y32.10. Information flow on pictorial, block, and signal flow diagrams shall be left to right and top to bottom. Fire control symbols shall conform to NAVSEA OP 1700.

5.1.1.3 Mechanically-reproduced copy preparation methods and computer output typesetting. Review manuscript, preliminary, and reproducible copy may be prepared by methods such as electric accounting machine, computer typesetting, punched card, magnetic tape (MTST or MTSC), paper tape, and shall conform to the requirements of 5.1, and the additional requirements of MIL-P-38790, for correction, mechanized assembly, and reproduction.

5.1.1.4 Microform compatibility. Microform compatibility of the SIB shall be as follows:

(a) Foldouts shall be prepared in the following format:
  (1) Foldout drawings shall be divided into 215 mm (8-1/2 inch) image area segments.
  (2) There shall be no drawing data, other than horizontal lines, placed closer than 3 mm (1/8 inch) from the image area limit.
  (3) Lines shall not be vertically displaced during the transition from one image area to the next.
(b) Halftone artwork shall not be used.
(c) Minimum printed size for text shall be 10 point. Nomenclature, callouts, tabular material and symbols on illustrations shall be upper case, with 8 point minimum printed size.
(d) Style and format shall be compatible for filming onto microfiche at a reduction ratio of 24:1 (98 images).

5.1.2 Binder, paper stock, and printing requirements. Cover and backbone stock, paper stock, and printing shall be in accordance with MIL-P-38790. Covers and backbones shall be printed on stock JCP R10 (891 pound type). The binder shall be constructed to permit ready insertion of revised pages.

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5.1.3 Security classification. Classified material shall be included in the SIB only where necessary and in the following manner:

(a) Where the extent of the classified material so dictates, or if reference to separate drawings, addenda, or other classified publications is not practicable and this material must be included in the basic publication, the entire manual, book or volume shall be classified.
(b) Where the classified material is minimized, such classified material shall be placed in an addendum and the basic publication shall not be classified. The addendum shall be a separate additional volume and shall be classified appropriately.
(c) Classified material shall be handled and marked in accordance with DOD 5220.22 Series.
(d) Classification of one volume of a publication shall not be the cause for classifying other volumes of the same publication.

5.1.3.1 Each technical publication prepared for the Government shall incorporate on the cover and title page the following distribution control statement, unless otherwise specified: "Distribution limited to DoD and DoD contractors only (operational use) (date statement applied). Other requests shall be referred to the Commander, Naval Sea Systems Command."

5.1.4 Warnings and cautions. Warnings are used in connection with operating or maintenance procedures, practices, conditions or statements, which if not strictly observed, could result in injury to or death of personnel. Cautions are used in connection with operating and maintenance procedures, practices, conditions, statements, which, if not strictly observed, could result in damage to, or destruction of, equipment or loss of mission effectiveness.

5.2 Front matter, covers, and backbone. Unless otherwise specified herein, requirements for front matter, covers, and backbone shall be in accordance with MIL-M-38784.

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5.2.1 Front matter. Front matter shall consist of the following contents:

(a) The front matter for each volume, supplement, or for part 1 of multi-part volumes, shall have the following contents in the order listed:
(1) Cover (including back cover and backbone),
(2) Publication change sheet (applicable only to changes),
(3) Title page,
(4) List of effective pages,
(5) Publication change record,
(6) Foreword,
(7) Table of contents,
(8) List of illustrations,
(9) List of tables,
(10) Safety summary, and
(11) List of abbreviations (Volume 1 only).

(b) The front matter for parts other than part 1 of multi-part volumes shall have the following contents in the order listed:
(1) Cover (including back cover and backbone).
(2) Title page.
(3) Table of contents.
(4) List of illustrations.
(5) List of tables (applicable to the part).
(6) List of abbreviations (Volume 1 only).

5.2.2 Front cover. Front cover shall contain the following:

(a) SIB identification number.
(b) Volume, part, supplement, and revision number, as applicable.
(c) The words "SHIP INFORMATION BOOK".
(d) SIB title and subtitle (volume title).
(e) NAVSEA seal.
(f) Ship name (or class) and identification number.
(g) Distribution statement.
(h) The words "PUBLISHED BY DIRECTION OF COMMANDER, NAVAL SEA SYSTEMS COMMAND".

If the SIB is classified, the front cover shall contain the security classification, espionage notice, and automatic time-phase downgrading and declassification information. When the SIB is a revision, or re-issue of an earlier publication, a supersedure notice shall be shown on the cover. If the manual is classified, the parenthetical note "(This cover is UNCLASSIFIED)" shall be placed beneath the classification marking at the bottom of the cover.

5.2.3 Back cover. The back cover, for each separately bound publication, shall contain the SIB identification number and, the applicable volume, part, supplement, and revision number. If the SIB is classified, the back cover shall also contain the security classification. When the SIB is less than 1 inch thick, the back cover shall also contain the title followed by the subtitle of the SIB.

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5.2.4 Backbone. Backbones shall be used for SIB that are 25 mm (1 inch) or more in thickness. The backbone shall contain the SIB identification number; the volume, part, supplement, and revision number, as applicable; the SIB title and subtitle; and the security classification, if applicable. This information shall appear vertically on the backbone of the SIB.

5.2.5 Title page. A title page shall be included in each SIB. It shall be a facsimile of the front cover, but shall also include the date of basic issue or revision. For a multipart volume, a blocked notice "SEE PART 1 OF THIS VOLUME FOR FRONT MATTER" shall be shown on the title pages of parts other than part 1. For a change, a new title page containing the change number and date of the change, shall be issued. For a change to a multipart volume, a new title page shall be issued for part 1 only. Title pages for parts other than part 1 shall not be changed.

5.2.6 Foreword. A foreword shall be included in each SIB supplement, volume, or in part 1 of a multipart volume. The foreword shall briefly describe the scope, content and structure of the entire SIB. The foreword shall include applicable classification guides for each subsystem documented. The foreword shall start as a right-hand page, immediately following the change record.

5.2.7 List of effective pages, change sheet, change record, table of contents, list of illustrations, list of tables, and safety summary. These items shall be prepared and included in the front matter in the order specified in 5.2.1.

5.2.8 List of abbreviations. The list of abbreviations shall include abbreviations used throughout the SIB.

5.2.9 Revisions. Changes shall be in the form of replacement pages corresponding to the list of effective pages (see 5.2.7). The revision date shall be indicated on the lower right hand corner of each revised page.

5.3 Structure of SIB. The SIB shall be arranged in chapters and sections and shall be in accordance with the data ordering document (see appendix). Volumes shall be divided into individually bound parts when required. Content of chapters and sections shall be in accordance with the volume arrangement provided in 5.3.3. Section and subsection content and location may vary within a particular chapter to accommodate necessary or unique data coverage, as specified by the contract.

5.3.1 Data level descriptions. Data presented shall be concise and to the depth necessary for supporting the objectives of the SIB (see 4.1).

5.3.2 Minimum data requirements. The text, illustrations and tables prescribed in this standard reflect minimum SIB requirements. Where the system configuration provides additional capabilities, operating modes or functions, other than covered herein, such additional coverage by text, illustrations and tabulated data shall be provided. Piping system diagrams shall include identification for all valves, either by words or by symbols.

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5.3.3 Volume arrangement.

5.3.3.1 Surface ships. Volumes 1 through 7 shall include the following descriptions and guidance:

(a) Volume 1 of the SIB shall include a general description of the ship's design characteristics and the major shipboard arrangements and systems. Descriptive material, lists, and illustrations, as applicable, shall be under the appropriate section number and title. Sections shall be in numerical sequence. Each section of the Ship Specification including all drawings and other documents developed therefrom, shall be carefully studied for applicable subject matter for this book. In addition, subjects such as docking and towing, which may not be covered in the ship specification but for which specific information is applicable and necessary for ship operation or maintenance, shall be included in the appropriate volume.

(b) Volumes 2 through 7 of the SIB shall describe ship systems, including design functions, operations, and limitations of each system, interrelationships between systems, trouble shooting, fault isolation techniques, system check-out, and routine systems maintenance. General or specific information contained in Combat Systems Technical Operations Manual (CSTOM), when prepared, the NSTM, and Equipment Technical Manuals, respectively, shall not be repeated in the SIB, except as necessary to fulfill the intent of the text. The following references, if of a permanent nature, shall be provided in the text, where applicable.
(1) Reference to appropriate drawings.
(2) Reference to drawing indices, books, schedules or other manuals such as damage control book, operational station book, schedule of watertight integrity tests and inspections, and equipment manuals.
(3) Cross references between volumes of the SIB, and between sections or chapters within a volume, shall be included only where necessary for continuity, to assist in complete understanding of the text and for avoidance of repetition.

(c) Volume 2 - Machinery plant. This volume shall consist of 2 parts.
(1) Part 1 - Propulsion plant general design and operation procedures. Part 1 shall acquaint and instruct the operators in the operation of the propulsion plant, shall describe the centralized control system, shall contain information and data in appropriate chapters and sections as outlined below, and shall conform, in general, to the sequence of the outline (as applicable to the type of propulsion plant):
a. List of references including principal equipment manuals or other technical manuals and drawings (using NAVSEA identifications) used in part 1 preparation, table of contents, list of illustrations, and legend of symbols.

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tests, inspections, cleaning requirements, adjustments and safety precautions, referred to in each chapter, shall be included at the end of each chapter, and shall refer to the page of the text where each is used. Tables shall be included showing the time required to flood and drain each damage control void and the time required to ballast and deballast each fuel and ballast tank or both, as applicable to expedite fueling-at-sea. The liquid loading instruction shall describe the manner in which protection against fragment damage can be maintained as ship fuel and aviation fuel are pumped down. The chapter on pumps shall be similar in format to the pump tables in section 503 of the Ship Specification and shall reference, in addition to the information contained in those tables, the applicable technical manual, compartment name and number of space containing each pump, title of system in which pump is located and the applicable SIB chapter number of the system. The chapter shall be categorized into the following subsections, as applicable: centrifugal, rotary, reciprocating, attached, hand and portable. The schedule for preparation of volume 2, part 2 shall permit issue of approved preliminary chapters at least 1 month prior to the operational check of the associated system. Part 2 shall include, but is not restricted to the following:

a. Auxiliary machinery plants:
   1. Auxiliary boilers,
   2. Refrigeration machinery,
   3. Air conditioning machinery,
   4. Steering system,
   5. Bow transfer system (where applicable) and,
   6. Fin stabilizer machinery (where applicable).

b. Air conditioning, ventilation and heating systems - Section 512
   1. Recirculating,
   2. Supply,
   3. Exhaust,
   4. Maintenance,
   5. Filter locations,
   6. Filter and duct cleaning instructions and,
   7. Collective Protection System (CPS) (where applicable).

c. Piping systems - One chapter for each type system as described in the noted section of the Ship Specification, as applicable.
   1. Steam - Section 253 and Section 517.
   2. Feed and condensate - Section 255 and/or Section 517.
   3. Sea water cooling - Section 256 and/or Section 521.
   4. Lubricating oil - Section 262.

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5. Fire main and sea water sprinkling - Section 521.
6. Plumbing drainage, deck drainage, etc. - Section 528.
7. Main, secondary and special drainage - Section 529.
8. Fresh water - Section 532.
9. Machinery and piping systems drainage - Section 534.
10. Ship fuel - Section 541.
11. Aviation fuel and gasoline - Section 542.
12. Compressed air - Section 551.
13. Compressed gas - Section 552.
14. Fire extinguishing - Section 555.
15. Special - Section 558.

d. Pumps - Section 503.

e. Environmental pollution control system - Section 593.
1. Sewage systems,
2. Waste water systems,
3. Solid waste system,
4. Oil pollution control system,
5. Air pollution control system,
6. Chemical and industrial waste system.

Except for the chapter on pumps described above, each chapter shall be subdivided into three sections as follows:

f. Section I - Description and principles of operation. This section shall be further subdivided into the three units described below:
1. Unit 1 - introduction. This unit shall cover the purpose, general description and major design characteristics of the system or independent auxiliary machinery plant, and shall include a diagrammatic, schematic drawing, or both.
2. Unit 2 - detailed description. This unit shall discuss the overall system or independent auxiliary machinery plant components, and its controls, instruments and its protective devices. Interrelationship between system components and the overall system or related companion system shall be included.
3. Unit 3 - principles of operation. This unit shall describe system or independent auxiliary machinery plant operations, list applicable precautions and discuss applicable operating stations. This unit shall convey information in an explanatory and descriptive manner as to what operations can be performed by or with the system or independent auxiliary machinery plant. Reference shall

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be made to corresponding operating instructions in section II for detailed step-by-step instructions, as applicable. Operations shall be discussed under the headings of Normal, Starting, Securing, Infrequent, Abnormal and Emergency. Abnormal operation shall be defined as an operation that results when specific equipment or components of a system or independent auxiliary machinery plant is not available for service. Emergency operation shall be defined as that operation performed as a result of a malfunction or maloperation of a system or component, the consequences of which prevent or seriously restrict the operation of the system or independent auxiliary machinery plant or jeopardizes the safety of ship, personnel or both.

g. Section II - Operating instructions. This section shall be further subdivided into the four units described below:

1. Unit 1 - Introduction. This unit shall simply describe the content of the section and reference the principles of operation described in section I.
2. Unit 2 - Initial conditions and precautions. This unit shall specify those initial conditions that provide a starting point for each operating instruction and shall caution and instruct the operator so that he fully understands the extent of the operation to be performed and its effect on system/independent auxiliary machinery plant conditions, and that he shall ascertain whether or not the instruction can be accomplished without conflicting with other operations. Initial conditions shall be based on valve line-up shown on system/independent auxiliary machinery plant diagrams for normal operations.
3. Unit 3 - Procedures. This unit shall cover detailed step-by-step procedures for performing the operations of the operating instruction under normal conditions. Where necessary, steps or instructions shall be emphasized by the use of "NOTE:" "CAUTION" or "WARNING" to insure safe procedures. Each operating instruction shall be provided with a minimum valve line-up table for performing the instruction or portion thereof. Procedures shall specify which steps may be

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performed in any sequence or in parallel. Procedures shall also include statements which permit the crew to omit certain steps under certain conditions, while still ensuring that crucial steps are performed in proper sequence. In addition, procedures shall be written to permit the crew to enter a procedure at any point, provided the conditions at that point are the same as they would have been had the procedure been performed starting from the initial condition stated in the procedure. Operating procedures in each chapter shall be based on the following format:

Initial conditions,
Precautions,
Summary check list (for immediate action on casualties only),
Operations flow diagrams (for complex evolutions only), and
Detailed procedure.

Notes and references to other operating instructions within the volume shall be minimized. Wherever possible, operating procedures shall be written to cover a single evolution and shall not contain extra steps intended to make the procedure cover more than one evolution.

4. Unit 4 - Back-up operation. This unit shall be used for those systems/independent auxiliary machinery plants that have back-up methods of operation and shall be addressed as being performed only when the procedures of Unit 3 are not practicable or possible. Back-up operations relating to the failure of any component or system that forms part of the HVAC portion of the Collective Protection System shall be exactly delineated, especially with respect to the loss of pressurization in a CPS zone.

h. Section III - Maintenance. This section shall be further subdivided into the three units described below:

1. Unit 1 - Principles of maintenance. This unit shall cover and discuss the importance proper maintenance has in retaining system/independent auxiliary machinery plant capability; discuss items which are generally applicable to a wide range of maintenance operations, such as cleanliness, lubrication and isolation criteria; reference sources of additional maintenance information; provide general precautions for performing maintenance; and shall define and briefly discuss maintenance instructions and maintenance procedures.

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each feeder emanating from the ship service switchboards, emergency switchboards, or load center switchboard.
Switchboard bus arrangement,
Generators (ship service, emergency, and special), motor generator sets, and solid state frequency changers (including electronics, aircraft servicing, interior communication, and weapons control),
Bus ties and bus feeders,
Shore power connections,
Line voltage regulators,
Voltage and frequency monitors,
Rectifiers,
No break power supplies,
Feedback circuits,
Transformers (lighting and interior communication),
Circuit breakers, (types, frame size, and element ratings),
Feeder designation with name and designation of load centers, power panels, transformers, or individual loads served, and
Generator prime mover types.

3. A diagram showing the normal, alternate, and emergency feeders to the following loads:
Steering gear,
Weapons systems,
Electronic systems,
Main drainage pumps and fire pumps, and
Interior communication switchboards.
In addition, bus ties and other interconnections shall be shown. For special types of ships, other loads shall be shown that are considered of the same order of importance as those listed above. This diagram shall show the loads, switchboards, and generators in block form with no interior connections or details such as circuit breakers or bus arrangements and shall show each of these elements at their approximate frame location. No effort shall be made to differentiate between different deck levels. The cables shall be shown in their actual location only insofar as they are to port, center, or starboard. For ships with three decks or platform levels or less (excluding superstructure), a wiring deck plan of the power systems shall be included. For all other ships, an isometric diagram of the power systems shall be provided.

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d. Switchboard - A description shall be given and illustrations included, where specified.
   1. Number and location.
   2. Use of each unit or section, for example, whether for generator, bus tie, or distribution.
   3. Any special features applying to switchboards, such as:
      Automatic operation of the emergency generating plant,
      Feedback arrangements, and
      Current controlled contactors.
   4. Precautions to prevent paralleling of emergency generators with each other or with other generators, or the requirements for paralleling, if so specified.
   5. Requirements for paralleling with shore power.

e. Central Control Console (where applicable) - A description shall be given and illustrations included, where specified.
   1. Location.
   2. Use of the unit (equipment or systems controlled or monitored).
   3. Any special features, such as:
      relinquinshing control to a local switchboard,
      automatic paralleling.
   4. Requirements for paralleling operating generators with shore power.
   5. Precautions to prevent loss of electrical plant if the Central Control Console is lost.

f. Protective devices - A description shall be given and the following illustrations included:
   1. A circuit breaker coordination chart shall be prepared showing the time-current characteristics of all the circuit breakers used in series from the generator to the load on a representative feeder. The bus tie circuit breakers plotted on log-log paper and shall cover the complete range from the long time delay pick-up value of current of the smallest breaker to the instantaneous trip point of the breaker with the highest setting.
   2. The short circuit current characteristics of the generator shall also be shown on this chart. Following are typical examples of the types, sizes, element ratings, and settings of the breaker circuits to be covered for selective tripping:
      Generator connections to switchboard,
      Bus tie,
      Feeder,
      Main and branch,
      Uses of fuses (on both power and lighting circuits),
      Design characteristics of controllers such as:
      Manual operation,
      Magnetic operation,
      Low voltage protection (LVP), and
      Low voltage release (LVR).

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g. Lighting systems - A description shall be given and illustrations included as applicable:
   1. For ships with three decks or platform levels or less (excluding superstructure), a wiring deck plan of the lighting systems shall be included.
   2. For all other ships, an isometric diagram of the lighting distribution system shall be included.
   3. Door switches and light traps for darkened ship operation.
   4. Red illumination, description of its operation and use.
   5. Relay and portable lanterns, description of its operation and use.
   6. Broad band blue illuminations, description of its operation and use.
   7. Navigation and signal light:
      Elementary wiring diagram,
      Outboard profile of the ship showing locations of the lights, and
      Operational procedures.
   8. Lights for night flight operations:
      Elementary wiring diagram,
      Equipment and light location, and
      Operational procedures.

h. Other systems and miscellaneous - A description shall be given and illustrations included:
   1. Shore power connections:
      Location,
      Reference to power system diagrams,
      Current capacity, and
      Hook-up procedures.
   2. Receptacles for portable connections for portable submersible pumps and portable welding sets:
      An illustration to show the locations and rating of outlets on the plan views of each deck on which outlets are located. In addition, an elevation view at the centerline shall be shown with all outlets as projected on the center plane. Sufficient compartmentation shall be shown to indicate accurately the location of outlets.
   3. Indication of length of cables supplied.
   4. How many cable types and in which repair station they are stowed.

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5. Arrangements for placing in simultaneous operation in each machinery room the maximum number of submersible pumps using the specified number and lengths of portable cables supplied. This shall be shown on both plan and elevation views.
6. A drawing showing the location of main wireways on all decks.
7. Casualty power system: A description of the purpose, scope, limitations, and proper use of this system. An isometric diagram of the complete casualty power system (showing terminals) shall be included.
8. Degaussing system: A general description of the system with instructions for operation. A sketch illustrating the approximate location of the coils.
9. Isolated receptacle circuits: A general description of this system including its purpose and need for maintaining high insulation resistance to ground.
10. Electrical schematic of lighting and outlet circuits and a table showing electrical feeders and breakers by compartment.

(2) Part 2 - General description of electric equipment and electrically-operated auxiliaries.

a. This part shall be divided into four chapters as follows:
   1. Chapter 1 - Power equipment,
   2. Chapter 2 - Electric-operated auxiliaries,
   3. Chapter 3 - Lighting equipment, and
   4. Chapter 4 - Miscellaneous equipment (test panels, storage batteries, instruments, and related equipment).

b. Data, as applicable, for each item of power and lighting equipment, exclusive of power equipment tabulated in chapter 2, shall be tabulated under the following headings in chapters 1, 3, and 4:
   1. Name of equipment.
   2. Name of manufacturer.
   3. Number of units.
   4. Location (designating number in accordance with the ship specification).
   5. Electrical data (brief).
   6. Controlling equipment (type and manufacturer).
   7. Reference to applicable technical manuals.

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APPENDIX

DATA REQUIREMENTS

10. DATA

10.1 Technical manuals. The requirement for technical manuals should be considered when this standard is applied on a contract. If technical manuals are required, Military specifications and standards which have been cleared and listed in DoD 5010.12-L (AMSDL) must be listed on a separate CDRL (DD Form 1423), included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

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